

2. Kinetic theory for an ideal gas gives an expression that can be written as:

$$pV = \frac{1}{3} nN_A m \overline{c^2}$$

(a) State the meaning of the terms: [2]

(i) m

(ii) $\overline{c^2}$

(b) Defining both symbols, explain what quantity is given by nN_A . [3]

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(c) The product of pressure and volume for an ideal gas may also be expressed as $pV = nRT$. Show in clear steps that the total translational kinetic energy of one mole of the gas is $\frac{3}{2} RT$. [4]

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- (d) An ideal monatomic gas, initially at a pressure of 115 kPa and temperature of 294 K, expands at constant pressure from a volume of $2.20 \times 10^{-3} \text{ m}^3$ to $2.60 \times 10^{-3} \text{ m}^3$. Calculate the change in the internal energy of the gas. [3]

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